



Bahir Dar University
Department of Geography and Environmental studies
Masters In Geo-Information Science

Spatial Database Design and Web GIS Development (Geis 633)

Web GIS project based Assignment

**Title: Development of Geospatial Data Infrastructure and Web-GIS
Portal for Bahirdar University GDTC using GeoServer.**

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1. PROBLEM DESCRIPTION AND OBJECTIVES

1.1 The Challenge at GDTC

The Geospatial Data and Technology Center (GDTC) at Bahir Dar University aims to serve as a hub for spatial data. However, the center currently stores spatial data in raw formats (Shapefiles, GeoTIFFs) across isolated drives. This creates a significant barrier for students and researchers who struggle to discover, access, and integrate datasets into their work. Additionally, although GDTC promises spatial data services as part of its community outreach, the absence of a service oriented geospatial infrastructure prevents fulfillment of this mission.

1.2 Project Objectives

- **Infrastructure Development:** Establish a robust, centralized Geospatial Data Infrastructure (GDI) that transitions GDTC from static file storage to dynamic data services.
- **Service-Oriented Architecture:** Implement prototype of Web GIS service portal for BDU-GDTC using GeoServer + PostGIS, enabling OGC-compliant data consumption (WMS/WFS).
- **Community Outreach:** Deliver a public-facing Web GIS portal where researchers can query, analyze, visualize, and download authoritative datasets.
- **Cloud Foundation:** Lay groundwork for a full “GeoServer Cloud” deployment, supporting future university-wide scaling.

2. SYSTEM ARCHITECTURE & COMPONENT SUMMARY

The infrastructure uses a **three-tier stack** that transforms raw spatial resources into actionable intelligence.

2.1 List of Software Tools and Their Description

All software used is open-source and was implemented on a **localhost** development environment.

- **PostgreSQL/PostGIS:** The core open-source object-relational database management system with advanced spatial capabilities for importing, storing, querying, and managing all geospatial data.
- **QGIS:** Used for initial data preparation, cleaning, geo-referencing, and importing data into the PostGIS database.

- **GeoServer:** The Java-based spatial data server application used to publish the spatial data from PostGIS as standard Web Services (WMS, WFS) for the web application.
- **XAMPP-Server:** A local Webserver like (WAMPP) platform used to host the web application components, pages on the localhost environment.
- **HTML, CSS, JavaScript:** The core web development languages used for the frontend user interface, ensuring interactivity and a responsive design.
- **Open layers/Leaflet JavaScript API:** A lightweight libraries used for displaying interactive web maps on the client-side browser.

3. KEY DATA REQUIREMENTS (INFRASTRUCTURE DEMONSTRATION)

To demonstrate the GDI's capacity to manage **vector and raster** datasets the following pilot datasets were imported to the PostGRE/PostGIS database:

1. **Administrative Layers:** Amhara Region Woredas, National Boundaries
2. **Infrastructure Layers:** Roads, Rivers
3. **Thematic Points:** Health Facility Locations
4. **Environmental Raster:** Land use land cover and Population Density

4. PRODUCTS ACCOMPLISHED AS A FINAL DELIVERABLE

4.1 Infrastructure Deliverables

- Fully normalized **PostGIS spatial database** hosting vector and raster datasets.
- Configured **GeoServer instance** for managing and publishing all datasets in **PostGIS** using OGC-compliant services (WMS/WFS).
- Dynamic **Web GIS Portal** (localhost/data catalogue/health portal) constructed using Leaflet.
- **Several operational spatial queries and analysis(viewer page):**
 - Search by health facility name/type
 - Identification of the *closest* health facility
 - Find closet facility
 - Find shortest route between two locations(user location & closest facility).

4.2 FINAL DELIVERABLES AND PRODUCTS

The infrastructure delivers the following integrated products as part of the GDTC Spatial Web GIS portal:

A. Web Modules

- **Central Portal Gate (index.html):** Landing/registration page
- **Unified Map Viewer (BasemapAndLayer.html/js):** Multi-basemap/multi data layers renderer.
- **JS based Interactive Viewer (viewer.html/js):**
 - Includes a *Spatial Query Analysis* panel
 - Allows facility filtering by type (Hospital, Clinic, etc.)
 - Features dynamic pop-ups and layer controls
- **Routing module (routing.html/js)** dedicated page with routing API key configured.

B. Analytical Engines

- **Proximity Search Tool:**
 - Radius-based search
 - User clicks a location → system buffers → queries GeoServer
- **Network Routing Module (routing.html):**
 - User defines Start/End points
 - Calculates *shortest path*, distance (km), and travel duration (min)
- **Closest Facility Engine:** Automatically identifies nearest health facility to a clicked point

5. DESCRIPTION OF WHETHER THE PROJECT OUTPUTS ARE LOGICALLY ORGANIZED

The project outputs are:

- **Logically structured**, with clear workflows connecting the backend (PostGIS/GeoServer) and frontend (Leaflet/openlayer interface).
- Designed with **intuitive navigation**, thematic layer grouping, and simple UX for non-GIS users.
- Hosted on a **contained localhost environment**, suitable for testing before campus-wide deployment.

6. DATA VISUALIZATION AND ACCESSIBILITY

- Layers can be styled using **QGIS SLDs and** imported into GeoServer.
- Intuitive Symbolology: (e.g. For health facilities)
- Health facility pop-ups immediately show Name, Type, etc.
- Multi-layer overlay(Regions, Weredas, Roads, health facilities)
- Base map Switcher (Satellite, Topographic, Street)
- Clear alert panels (e.g., when queries return zero results)

- Transparent administrative boundaries to avoid obscuring point data
- Symbolization Icons for each health facility type (Hospital, Health Center, Health Post, and Clinic).

7.3 Site Map

- **index.html** → Entry point
- **viewer.html** → Main Viewer
- **routing.html** → Routing Engine
- **BasemapAndLayer.html** → Base map and Layer Browser
- **AdvancedGeoServer.html** → link to GeoServer page, data catalogue, GeoServer layer viewer, CQL based querying, filtering and analysis.

8. DEMONSTRATION OF SYSTEM

The prototype GDTC Geospatial Infrastructure demonstrates real-world problem-solving ability: it is applied on localhost accessed via:

- GeoServer (<http://localhost:8080/geoserver/web/?0>) and
- XAMPP server (<http://localhost/index.html>).

Typical ADMINISTRATIVE WORKFLOW (BACKEND MANAGEMENT)

1. **Ingest:** Raw datasets into PostGIS
2. **Configure:** GeoServer with PostGIS, organize data into Workspaces, stores, layers, DB connections, and several server administration activities.
3. **Publish:** Layers activated as WMS/WFS for public consumption
4. **Style:** SLD-based design(QGIS styles can be imported to GeoServer).

RESEARCHER & PUBLIC USER WORKFLOW (FRONTEND ACCESS)

1. **Discovery:** index.html
2. **Visualization via Leaflet pages :** Viewer displays multiple custom menus and layers
3. **Analysis:** "Find Nearest" and other tools execute queries
4. **Visualization via GeoServer built-in pages :** advanced layer view, query features.
5. **Data access and extraction:** Users can filter and download data in several formats GeoTIFF, GeoJSON, jpeg via WFS.

9. CONCLUSION & FUTURE DIRECTION TO CLOUD SERVICE

The research successfully achieved the objectives; It project can be scaled and used to transition BDU-GDTC from fragmented datasets to a **modern, cloud-ready Geospatial Infrastructure**, eliminating licensing barriers and establishing a university-wide Spatial Cloud Service. Future works and plans include integrating “GeoServer Cloud” which allows to scale up and provide full cloud service—a mission envisioned by BDU-GDTC.

Reference

- 1) GeoServer <https://geoserver.org/>
- 2) GeoServer: Complete Tutorial.
https://www.youtube.com/watch?v=FLATWBAhwS0&list=PLumZ7YQFbxIyBx3_zzbUHUxmIAKgIPRcF